Sanath N Upadhyaya

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PROFESSIONAL SUMMARY

Software engineer specializing in applied and post-quantum cryptography (ML-KEM / FIPS 203, ZK proofs), distributed systems, and secure infrastructure. Experience building automated testing frameworks for large-scale database systems, with graduate research in lattice-based cryptographic proof systems.

EDUCATION

Master of Science in Computer Science — Aug 2025 - Present
Northeastern University, Boston, MA — GPA: 4.0
Coursework: Programming Design Paradigm, Reinforcement Learning, Algorithms, Database Systems
Graduate research: lattice-based cryptographic proof systems (zero-knowledge proofs), Khoury Cryptography & Privacy Group.

Bachelor of Engineering in Electronics and Communication Engineering — 2019 - 2023
BMS College of Engineering, Bengaluru, India — GPA: 9.15 / 10
Coursework: Operating Systems, Machine Learning, Internet of Things, Number Theory

WORK EXPERIENCE

Software Development Engineer Intern | Amazon Web Services, Palo Alto, CA — May 2026 - Aug 2026

- Built an automated integration-testing framework for Amazon Aurora (MySQL-compatible) from the ground up, generating and validating third-party workloads against distributed database clusters.
- Implemented differential-testing pipelines against reference databases to surface correctness discrepancies across query-execution paths.
- Designed agent-orchestration workflows to automate test generation, execution, and validation, cutting manual effort in large-scale compatibility testing.

Cloud Developer | Hewlett Packard Enterprise, Bengaluru, India — Aug 2023 - Aug 2025

- *Security Engineering:* Designed and operated authentication and access-control systems (OAuth2, Keycloak), migrating legacy LDAP to centralized identity (GreenLake IAM, Okta) and driving adoption across internal services.
- *Infrastructure Automation:* Automated OpenShift cluster provisioning with Python and REST APIs, ensuring consistent security baseline enforcement across deployments and reducing setup time by 30%.
- *Zero-Trust Architecture:* Implemented zero-trust service communication using Istio and SPIRE, enabling identity-based access control and improving visibility into service interactions.
- Monitored system metrics and logs using Prometheus and related tooling to identify anomalies and performance issues in distributed services.

Software Engineering Intern | Hewlett Packard Enterprise, Bengaluru, India — Jan 2023 - Jul 2023

- Defined and enforced security policies for Ezmeral Runtime Engine (ERE), integrating OAuth2-proxy for access control across multi-tenant environments.
- Enabled OpenShift support for the HPE Unified Analytics (UA) tool and optimized Prometheus integration for improved performance.

PROJECTS

Post-Quantum & ZK Proof Benchmarking Framework | Github — **Rust, ml-kem, arkworks, Docker, Kubernetes**

- Designed HTTP microservices in Rust wrapping ML-KEM (NIST FIPS 203) key encapsulation and arkworks R1CS Groth16 zero-knowledge proof generation.
- Benchmarked ML-KEM-512, ML-KEM-768, and ML-KEM-1024 parameter sets against Groth16 proving; measured ~17x throughput gap, quantifying the performance cost of cryptographic verifiability versus post-quantum key exchange.
- Containerized services with Kubernetes manifests for horizontal scaling experiments, simulating throughput under concurrent load to evaluate PQC readiness for production workloads.

End-to-End Encryption Engine with Double Ratchet | Github — **Java, Java Cryptography Extension (JCE),**

- Implemented Double Ratchet–based E2E encrypted async messaging in Java, achieving forward secrecy and post-compromise security.
- Composed public-key (ECDH) and symmetric (AES-256-GCM, HKDF, HMAC-SHA256) primitives in a ratcheting construction resilient to server compromise and session-key exposure.

Akshara Mantapa: Infinite Kannada Text Library | Github | Web-Demo — **Rust, WebAssembly, SvelteKit**

- Architected a deterministic text-generation and search engine in Rust (Axum + WebAssembly) handling a 57k-symbol Unicode grapheme space, designed for correctness under adversarial input segmentation.

TECHNICAL SKILLS

Languages : Python, Rust, Go, C++, Java.
Cloud Technologies: AWS, GCP, Azure, Kubernetes, OpenShift, Jenkins, Docker, Istio, SPIRE, Rancher, Prometheus, Ansible, Terraform
Security & Systems : Authentication, OAuth2, Key/Secrets Management, Zero-Trust Architecture, IAM, Secure Communication
Certifications : Certified Kubernetes Administrator (**CKA**), Certified Kubernetes Security Specialist (**CKS**)
Testing: pytest, Selenium, differential testing, integration and end-to-end frameworks
Applied Cryptography: AES-GCM, ECDH, HKDF, HMAC, ML-KEM (FIPS 203), Groth16 ZK, Double Ratchet